

Questions for the Audience

What are the two Goals of Open-Set Recognition?

- Correctly classify samples from known classes
- Assign no class to samples from unknown classes

Which Integrated Approaches to Handle Unknown Samples exist?

- Include an additional unknown/background/garbage class
- Reduce the confidence for all known classes

Which Methods make use of a Pre-Trained Network for OSR?

- Simple thresholding of softmax probabilities or logits
- OpenMax estimates probability of unknown
- GHOST reduces confidence for unknown samples

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Which Variations of Auto-Encoders do you Know?

- Simple two or four-layer auto-associative networks
- Convolutional auto-encoders with sparsity constraints
- Denoising auto-encoders, including diffusion models
- Variational auto-encoders (for data generation)
- Foundation models (see Lecture 10)

What are the two Parts of Generative Adversarial Networks?

- The generator to generate samples from noise
- The discriminator to distinguish real from created samples